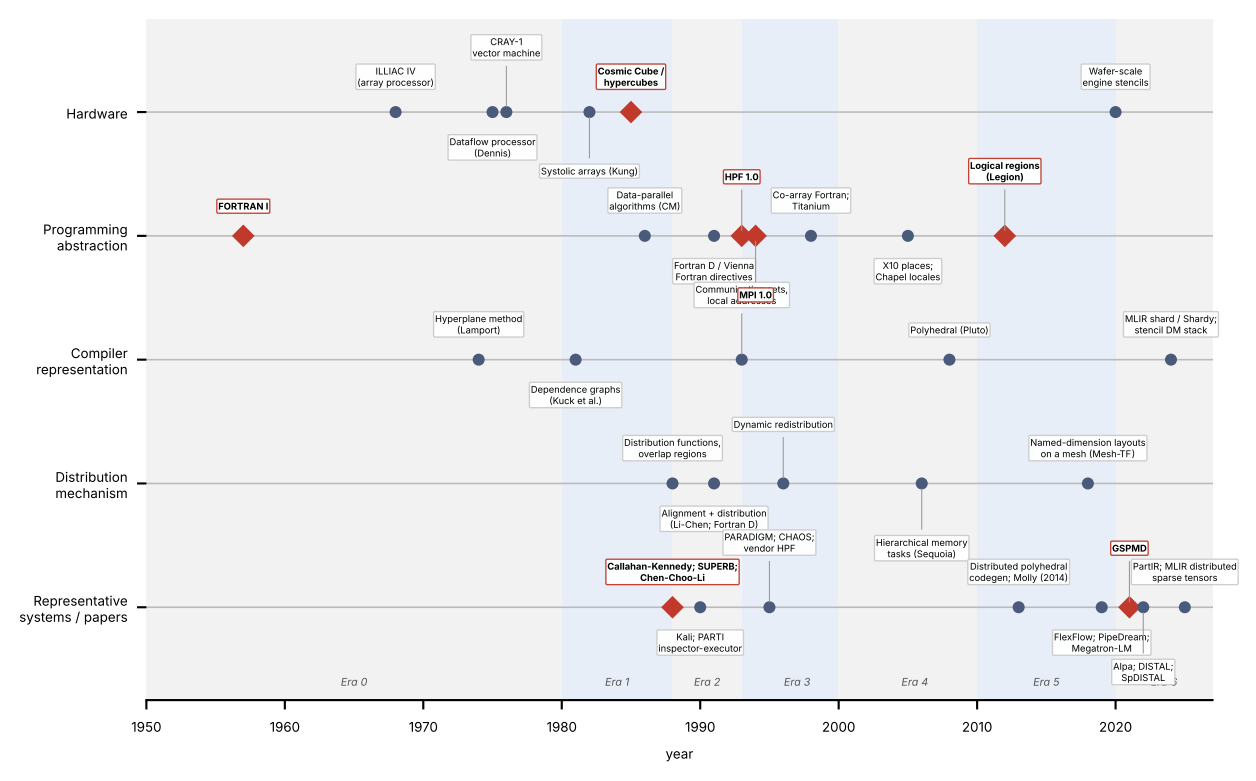
Hardware
Programming abstraction
Compiler representation
Distribution mechanism
Representative systems / papers
ILLIAC IV (array processor)
CRAY-1 vector machine
Cosmic Cube / hypercubes
Wafer-scale engine stencils
Dataflow processor (Dennis)
Systolic arrays (Kung)
FORTRAN I
Data-parallel algorithms (CM)
HPF 1.0
Co-array Fortran; Titanium
Logical regions (Legion)
Fortran D / Vienna Fortran directives
X10 places; Chapel locales
MPI 1.0
Hyperplane method (Lamport)
Polyhedral (Pluto)
MLIR shard / Shardy; stencil DM stack
Dependence graphs (Kuck et al.)
Dynamic redistribution
Distribution functions, overlap regions
Named-dimension layouts on a mesh (Mesh-TF)
Alignment + distribution (Li-Chen; Fortran D)
PARADIGM; CHAOS; vendor HPF
Hierarchical memory tasks (Sequoia)
GSPMD
Callahan-Kennedy; SUPERB; Chen-Choo-Li
Distributed polyhedral codegen; Molly (2014)
PartIR; MLIR distributed sparse tensors
Kali; PARTI inspector-executor
FlexFlow; PipeDream; Megatron-LM
Alpa; DISTAL; SpDISTAL
Era 0
Era 1
Era 2
Era 3
Era 4
Era 5
1950
1960
1970
1980
1990
2000
2010
2020
year